

GO-CAMPUS

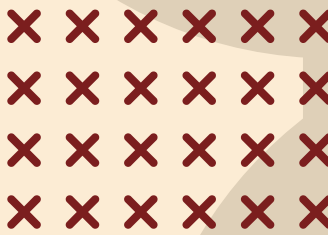
DYNAMIC CAMPUS SHUTTLE TRACKING AND LOAD MONITORING SYSTEM

Presented by: Thayalan A/L Ramasamy

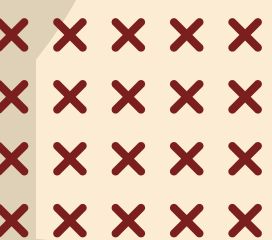
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INTRODUCTION



- Go-Campus is an IoT-based system designed to provide real-time tracking of campus shuttle movement and passenger load.
- The system uses a GPS module to continuously track the shuttle's location and sensors to detect the number of passengers entering and exiting the vehicle.
- With Go-Campus, students can check shuttle availability, location, and occupancy status before arriving at the bus stop, reducing unnecessary waiting time and avoiding full shuttles.
- The system supports efficient campus mobility by improving travel planning and enhancing the overall student commuting experience.
- Go-Campus combines hardware (ESP32/NodeMCU, GPS, sensors) and software (cloud storage, dashboard interface) to deliver a complete smart transportation solution.



PROBLEM STATEMENT

- Students currently have no access to real-time shuttle location or arrival updates, causing uncertainty about shuttle schedules. As a result, students often spend excessive time waiting at shuttle stops without knowing when the shuttle will arrive, leading to frustration and wasted time especially during peak academic hours.
- The campus shuttle system does not provide real-time information on shuttle occupancy, meaning students only discover that a shuttle is full after it arrives. This results in overcrowding, discomfort, and students being forced to wait for the next shuttle, causing delays and affecting punctuality for classes or appointments.
- Due to the lack of automatic data collection on shuttle usage such as real-time passenger count and shuttle movement administrators are unable to optimize shuttle routes or scheduling efficiently. This leads to poor resource allocation, inefficient route planning, and inconsistent shuttle frequency, reducing overall service effectiveness.

PROJECT OBJECTIVES

The main objective of this project is to create an IoT-based system that improves the efficiency of campus shuttle services. The system will show real-time shuttle location and passenger load so students can avoid long waiting times and overcrowded shuttles. By combining sensors, GPS tracking, and a cloud dashboard, the system helps students plan their travel better and provides useful data for campus management to improve shuttle schedules and routes.

- To study the current campus shuttle operations and identify issues related to waiting time, overcrowding, and lack of real-time information.
- To develop an IoT-based system that integrates GPS tracking and passenger counting sensors, and transmits real-time shuttle location and occupancy data to a cloud platform accessible through a mobile or web dashboard.
- To evaluate the system's accuracy, performance, and usability by testing real-time data transmission, dashboard responsiveness, and effectiveness in reducing waiting time and overcrowding.

USER GOALS

- User scope describes who will use the system, what their roles are, and how they interact with the system.
- It defines the target users and their access level or functions within the system.

User Type	Role	Functions / Access
Admin (System Manager)	Oversees and manages the entire Go-Campus system.	Manages shuttle information (add / update shuttle details).Monitors real-time shuttle routes and passenger load. Manages user accounts and permissions. Generates reports for shuttle usage and passenger statistics. Adjusts shuttle schedules or routes based on collected data.
Shuttle Driver / Shuttle Operator	Operates the shuttle equipped with sensors and IoT module.	Ensures the IoT device is active during operation.Can view current shuttle route or route updates. Notifies admin in case of system/device issues (optional feature).
Student / Public User (Passenger)	Main end-user who views information to plan trips.	Checks real-time shuttle location on the map.Views shuttle occupancy status (available seats / full).Plans travel by choosing the best shuttle based on location and load.No editing or administrative privileges.

PROJECT SCOPE

- The project covers the development of an IoT-based system to track campus shuttle location and passenger load in real time.
- The system includes hardware installation on the shuttle (GPS module + passenger counting sensors + ESP32/NodeMCU).
- Shuttle location and passenger count data are sent to a cloud platform (e.g., Firebase / ThingSpeak).
- A web or mobile dashboard will be developed to display real time shuttle position and passenger occupancy.
- Students can only view information — no modifications allowed.
- Admin (system manager) can register shuttles , monitor usage data and manage system settings
- The system does not include payment system , ticket booking.
- The scope is limited to campus environment and Wi-Fi/cloud connectivity availability.

LITERATURE REVIEW - EXISTING SYSTEMS

- The integration of the Internet of Things (IoT) with transportation systems has been widely researched to improve mobility, reduce waiting time, and enhance user experience. Many studies highlight the use of GPS tracking, cloud connectivity, and real-time data visualization to optimize vehicle routing and provide live location updates to users. IoT-based passenger monitoring has also been explored using sensors to detect occupancy levels, helping reduce overcrowding and improve travel planning.
- The Go-Campus project builds upon these concepts by combining GPS technology, passenger counting sensors, and a cloud platform to monitor campus shuttle movement and load in real time. The system uses an ESP32/NodeMCU microcontroller to collect data from the sensors and transmit shuttle information to a web or mobile dashboard, allowing students to view shuttle availability, location, and occupancy before traveling.

LITERATURE REVIEW - EXISTING SYSTEMS

No.	Author / Year	Title	Algorithm/ Technnique Used	Advantages	Disadvantages
1	P. Subramaniam, et al. (2019)	Passenger and Luggage Weight Monitoring System for Public Transport	Rule-Based Sensor Calculation (Load-cell weight sensing linked to microcontroller logic)	- Monitors vehicle load capacity in real time. - Helps prevent overloading and improves passenger safety. - Automated data collection with minimal human intervention.	- Only measures total weight, not exact passenger count. - Does not track vehicle location. - Accuracy may vary depending on sensor calibration.
2	Sujitha K., et al. (2020)	IoT Based Bus Arrival Time Prediction Using Artificial Neural Network for Smart Public Transport System (SPTS)	Artificial Neural Network (ANN)	- Predicts bus arrival time using machine learning. - Reduces uncertainty and waiting time for passengers. - Real-time bus tracking improves user experience.	- No load or seat availability monitoring. - Requires stable internet and accurate datasets for training. - Complex implementation compared to rule-based tracking.

LITERATURE REVIEW - EXISTING SYSTEMS

3	Karthik S., et al. (2023)	Bus Tracking and Arrival Prediction System	Rule-Based GPS Tracking (Location update logic via GPS → Server → App)	- Allows passengers to track bus live location. - User-friendly mobile/web dashboard. - Improves efficiency and reduces waiting time.	- No passenger occupancy detection. - Tracking accuracy depends on GPS signal strength. - Limited decision-making analytics for campus management.
4	Mohamed A., et al. (2023)	Passenger Occupancy Estimation in Vehicles: A Review of Methods	Computer Vision / Machine Learning / IR Sensors	- Provides insights on multiple techniques for passenger counting. - Helps identify most efficient detection approach. - Supports optimization of public transportation systems.	- Some methods are costly (e.g., camera-based systems). - Privacy concerns when using video-based monitoring. - Sensor-based systems may miscount during peak movement.

LITERATURE REVIEW - EXISTING SYSTEMS

5	Zhang, H., Li, W., & Wang, Y. (2023)	Research on Dynamic Scheduling and Route Optimization Strategy for Flex-Route Transit	Dynamic scheduling model + optimization algorithms for real-time routing	- Real-time flexible routing. - Can adjust to passenger demand and traffic conditions. - Applicable to campus shuttle scenarios.	- Implementation details limited. - Requires IoT/real-time data infrastructure.
6	Lin, J., Chen, L., & Wu, X. (2022)	Multicriteria Route Planning for In-Operation Mass Transit Systems	Heuristic spanning tree algorithms for multi-criteria route planning (flow, fuel, transfers, welfare)	- Optimizes multiple objectives simultaneously. - Useful for balancing efficiency, passenger convenience, and operational cost.	- Focus on large-scale transit; may need adaptation for small shuttle fleets. - Computationally heavier than simple heuristics.

LITERATURE REVIEW - EXISTING SYSTEMS

7	Zhao, P., Liu, Q., & Wang, R. (2022)	Transit Dynamic Operation Optimization Using Stop-Skipping Control and Local Route Optimization	Mixed strategy: stop-skipping + local route optimization to reduce passenger wait & operational costs	- Reduces travel and operational costs. - Dynamic adaptation to changing passenger demand. - Can be tailored for single-line campus shuttles.	- Limited to single-line routes. - Requires real-time passenger data for effectiveness.
8	Kim, S., Park, H., & Lee, D. (2023)	Service Management of Employee Shuttle Service Under Dynamic Demand	Dynamic fleet management model (optimization under variable demand)	- Handles variable demand. - Optimizes shuttle allocation and scheduling. - Directly relevant to campus shuttle systems.	- Focused on employee shuttles; adaptation may be needed for students. - Needs accurate demand prediction for efficiency

LITERATURE REVIEW - EXISTING SYSTEMS

9	Zhang, Y., Chen, F., & Liu, H. (2022)	Integrated Multi-Objective Optimization for Dynamic Airport Shuttle Bus Services	Multi-objective mixed-integer linear programming (MILP) for route, frequency, and station choice	- Multi-objective optimization addresses multiple constraints. - Dynamic adaptation to time-varying demand. - Can inspire approaches for campus shuttle optimization.	- Computationally intensive. - Designed for airport-scale; may need simplification for campus-scale shuttle routes.
10	Sun, Q., Hu, D. W., Chien, S., et al. (2022)	Dynamic Bus Routing Optimization for Demand-Responsive Feeder Transit Considering Stochastic Bus Arrival Time	Mixed-integer optimization + heuristic (Genetic Algorithm + neighbourhood search)	- Handles real-time demand dynamically. - Models stochastic bus arrival times, improving realism. - Reduces passenger waiting time and operational costs.	- Requires detailed arrival time and demand data. - Heuristic may not guarantee global optimality. - Designed for feeder transit; may need adaptation for campus shuttle.

SCREENSHOTS - EXISTED SYSTEMS

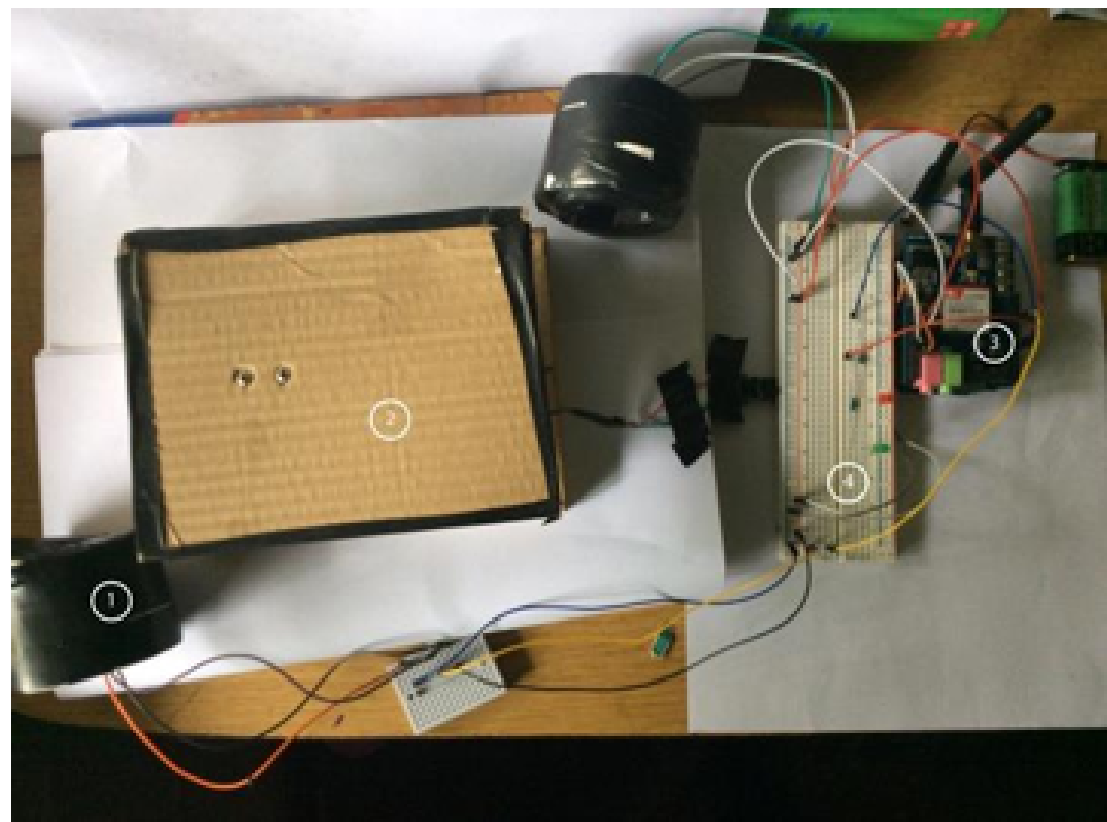
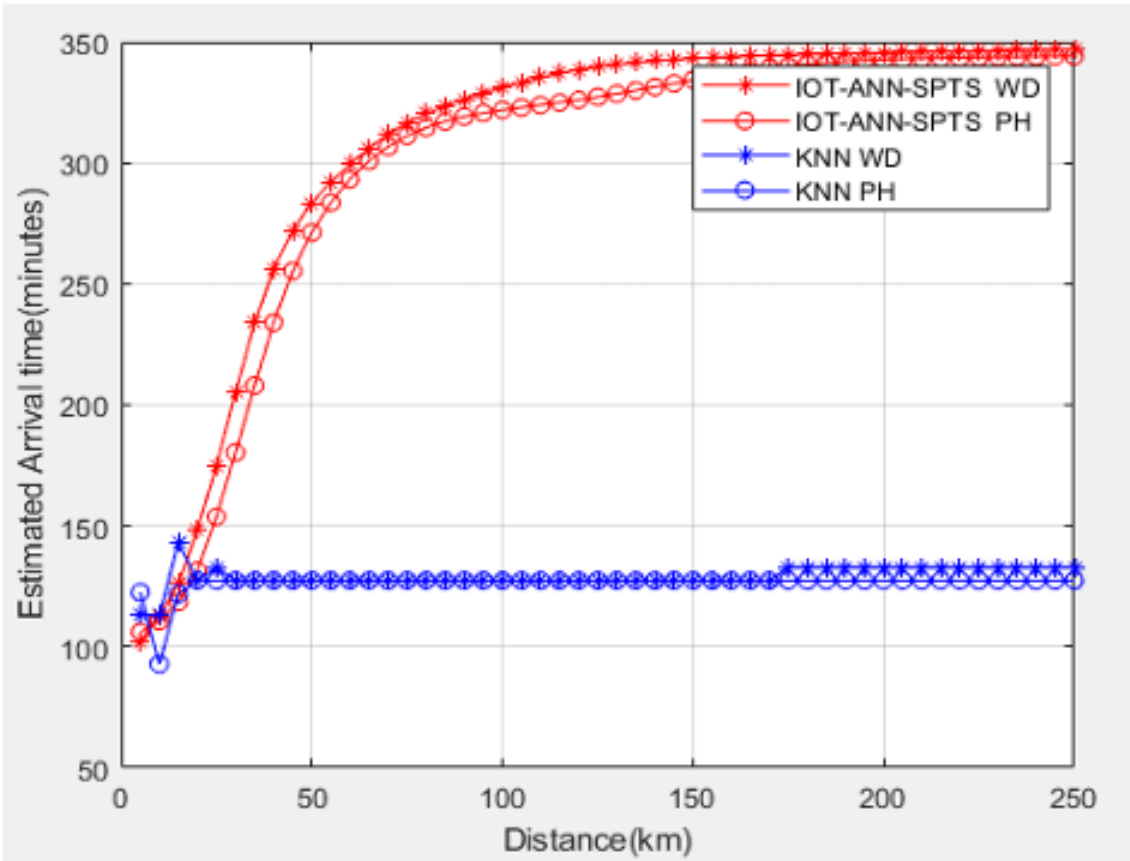


Table 2. comparison for actual and predicted time for KNN [17] prediction

Distance (km)	1	5	10	15	20	25	30
Observed arrival time(minutes)	98.60	101.09	103.65	106.20	108.75	111.31	113.86
Estimated arrival time (minutes)	116.23	116.23	116.23	113.79	113.79	113.79	113.79

Table 3. Comparison for actual and predicted time for ANN

Distance (km)	1	5	10	15	20	25	30
Observed arrival time(minutes)	98.60	101.09	103.65	106.20	108.75	111.31	113.86
Estimated arrival time (minutes)	98.53	101.08	103.65	106.213	108.76	111.31	113.86



PROJECT OVERVIEW

- Internet of Things (IoT): Enables communication between sensors, GPS modules, and the cloud to support real-time tracking and monitoring of shuttle activity.
- GPS Technology: Provides continuous tracking of shuttle location and movement data, allowing users to know the exact position on a digital map.
- Infrared / Load Sensors: Detect and count passengers entering or exiting the shuttle to determine occupancy and seat availability.
- Cloud Computing: Stores and processes real-time shuttle and passenger data, enabling remote dashboard access for students and administrators.
- Web/Mobile Dashboard (UI/UX): Displays shuttle route, live location, and passenger load, allowing users to plan travel efficiently before reaching the shuttle stop.
- The proposed system uses a rule-based passenger counting algorithm combined with MQTT for real-time data transmission. The rule-based logic updates the passenger count automatically whenever the IR sensor detects entry or exit activity, ensuring accurate occupancy monitoring without requiring user interaction. MQTT is used to send sensor and GPS data to the cloud efficiently, offering low latency and low bandwidth usage compared to traditional HTTP methods. This algorithm is more practical than camera-based or machine-learning approaches because it is cost-effective, privacy-safe, and provides real-time responsiveness, making it ideal for campus shuttle tracking and load monitoring.

GANTT CHART

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GO CAMPUS GANTT CHART															
NO	ACTIVITIES	WEEK													
		W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
1	Topic Discussion and Selection														
2	Topic Submission														
3	Proposal Writing (Introduction)														
4	Proposal Writing (Problem Statement,Objective and Scope)														
5	Literature Review														
6	Project Proposal Presentation (Proposal Defend)														
7	Project Proposal Correction														
8	Methodology Writing														
9	Prototype Design														
10	Report Writing														
11	Project Proposal Final Presentation														
12	Project Proposal Report Submission														

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PROS AND LIMITATION OF SELECTED SYSTEMS AND THEORIES

Aspect: Real-Time Shuttle Tracking

Pros:

- Students can view shuttle location live on a map.
- Reduces unnecessary waiting time at shuttle stops.
- Improves planning and time management.

Cons:

- GPS accuracy may decrease in bad weather or near tall buildings.
- Requires stable internet connection for real-time updates.

Aspect: Cloud & Dashboard System

Pros:

- Students and admins can access data from anywhere.
- Dashboard provides visual and user-friendly interface.
- Enables live monitoring without manual checking.

Cons:

- Cloud services may require ongoing maintenance.
- System performance depends on internet speed.

Aspect: Passenger Load Monitoring

Pros:

- Shows shuttle occupancy (available seats / full).
- Prevents overcrowding and improves passenger comfort.
- Helps shuttle management identify peak usage times.

Cons:

- Sensor accuracy may be affected when multiple students board at once.
- Requires proper installation and calibration in the shuttle.

Aspect: Hardware (ESP32 / GPS / Sensors)

Pros:

- Uses affordable and widely available IoT components.
- Easy to prototype and integrate with cloud platforms.

Cons:

- Requires stable power source when installed in shuttle.
- Hardware may experience physical wear over time.

EXPECTED OUTCOMES

- A functional prototype of the Dynamic Campus Shuttle Tracking and Load Monitoring System (Go-Campus) will be successfully developed.
- Students will be able to view the real-time location of campus shuttles through a mobile or web interface.
- The system will help reduce waiting time and uncertainty at shuttle stops by providing live updates.
- A passenger load monitoring feature will show the shuttle's current occupancy level (number of available seats).
- Students can make informed decisions on whether to wait or board another shuttle based on seat availability.
- Campus management will have access to a cloud dashboard to monitor shuttle routes, passenger load, and real-time usage data.
- The system will support data-driven decision making, enabling better route planning and shuttle scheduling.
- Overall, the system will create a more efficient, user-friendly, and transparent shuttle service.
- Improves resource allocation and reduces overcrowding during peak hours.

METHODOLOGY

1. Requirements Planning

- Identify system goals and scope
- Define student, driver, and admin roles
- Prioritize core features

2. System Design

- Design system architecture
- Prepare DFDs, ERD, and flowcharts
- Plan mobile app interfaces

3. Development

- Implement IoT data collection
- Develop mobile app features
- Apply alert and status logic

4. Testing

- Test individual modules
- Perform integration testing
- Validate alert conditions

5. Deployment

- Deploy prototype system
- Run using simulated data
- Prepare for demonstration

6. Review & Feedback

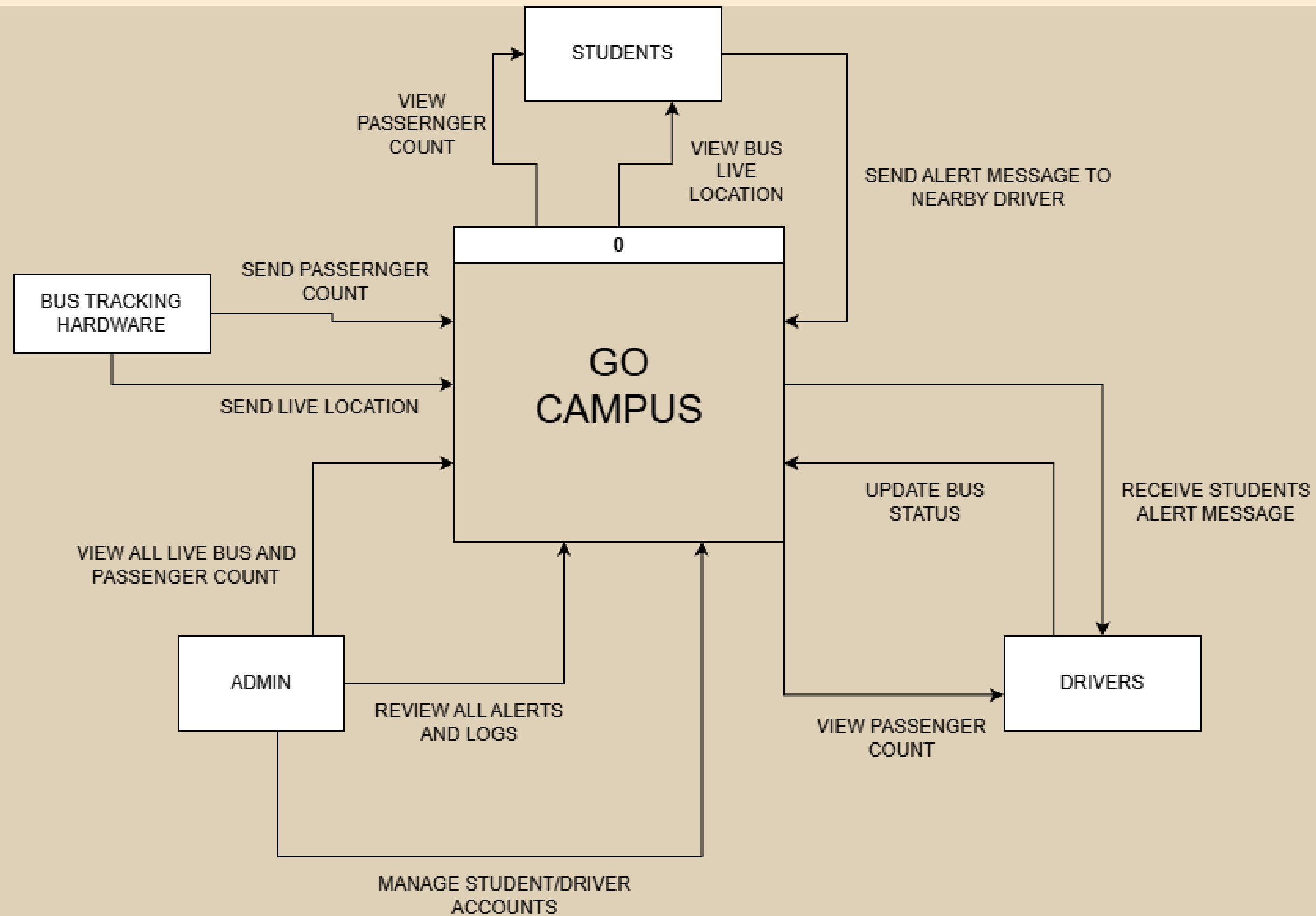
- Collect supervisor feedback
- Evaluate system performance
- Identify improvements

7. Maintenance

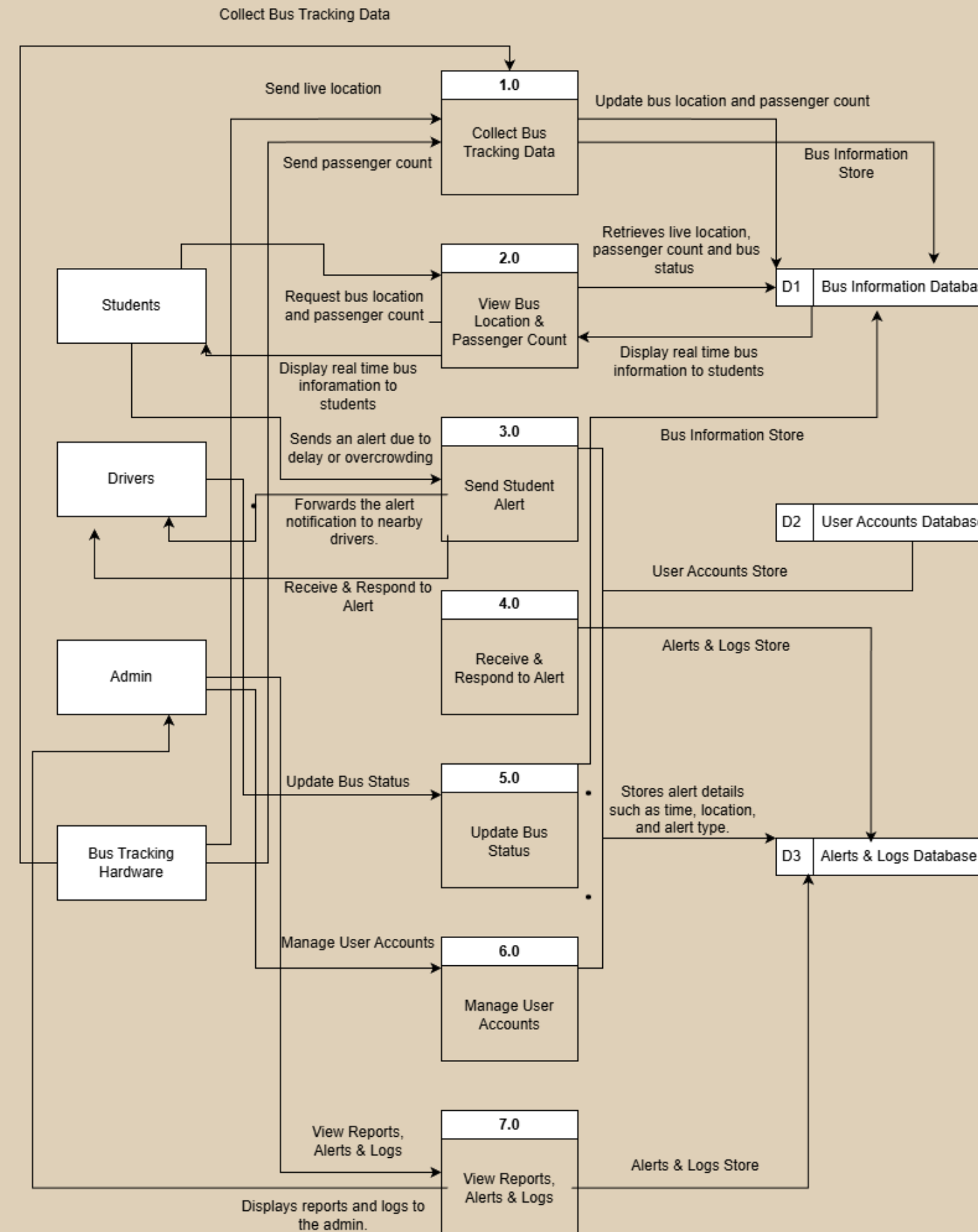
- Refine existing features
- Optimize system logic
- Support future enhancements



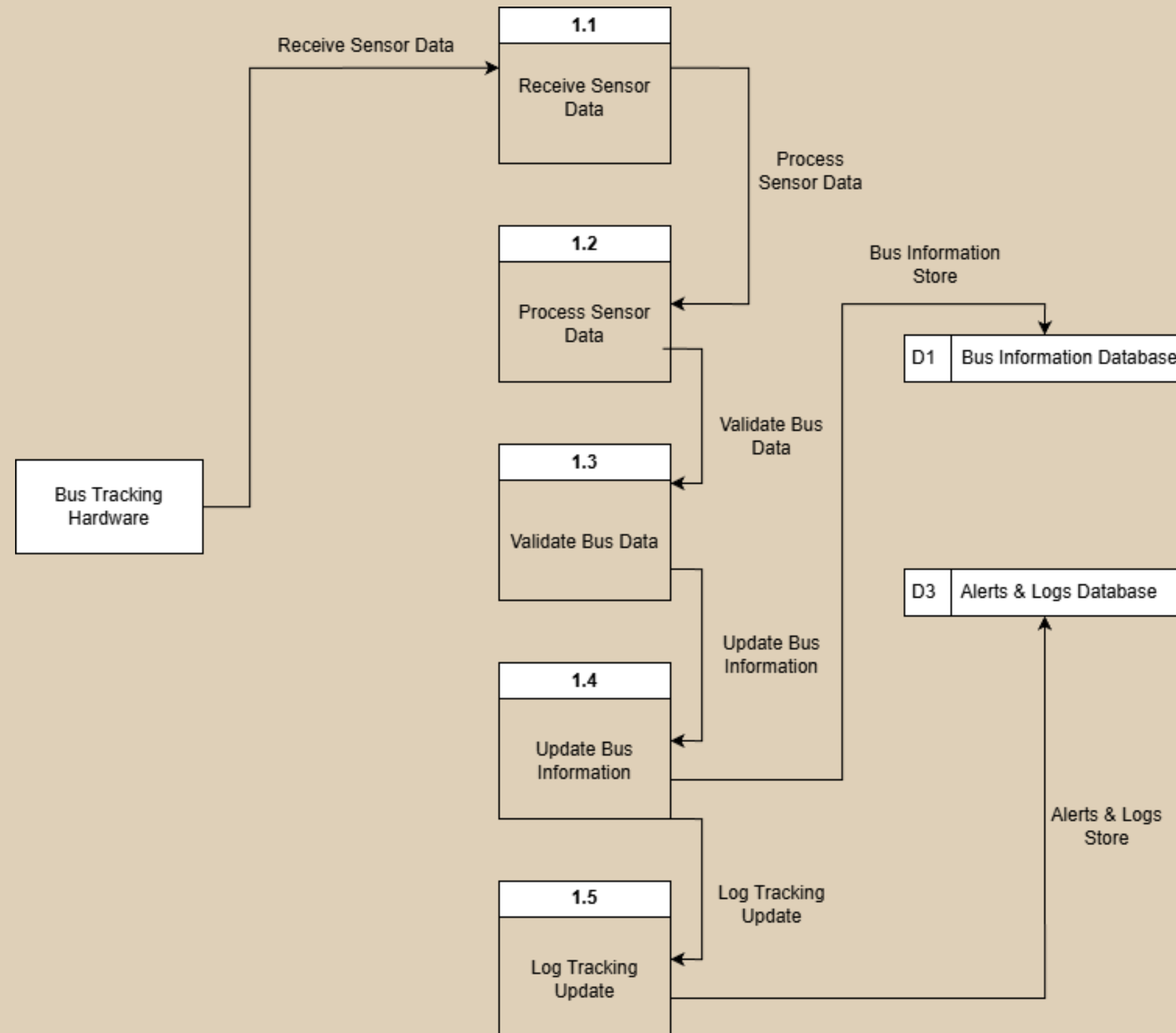
CONTEXT DIAGRAM



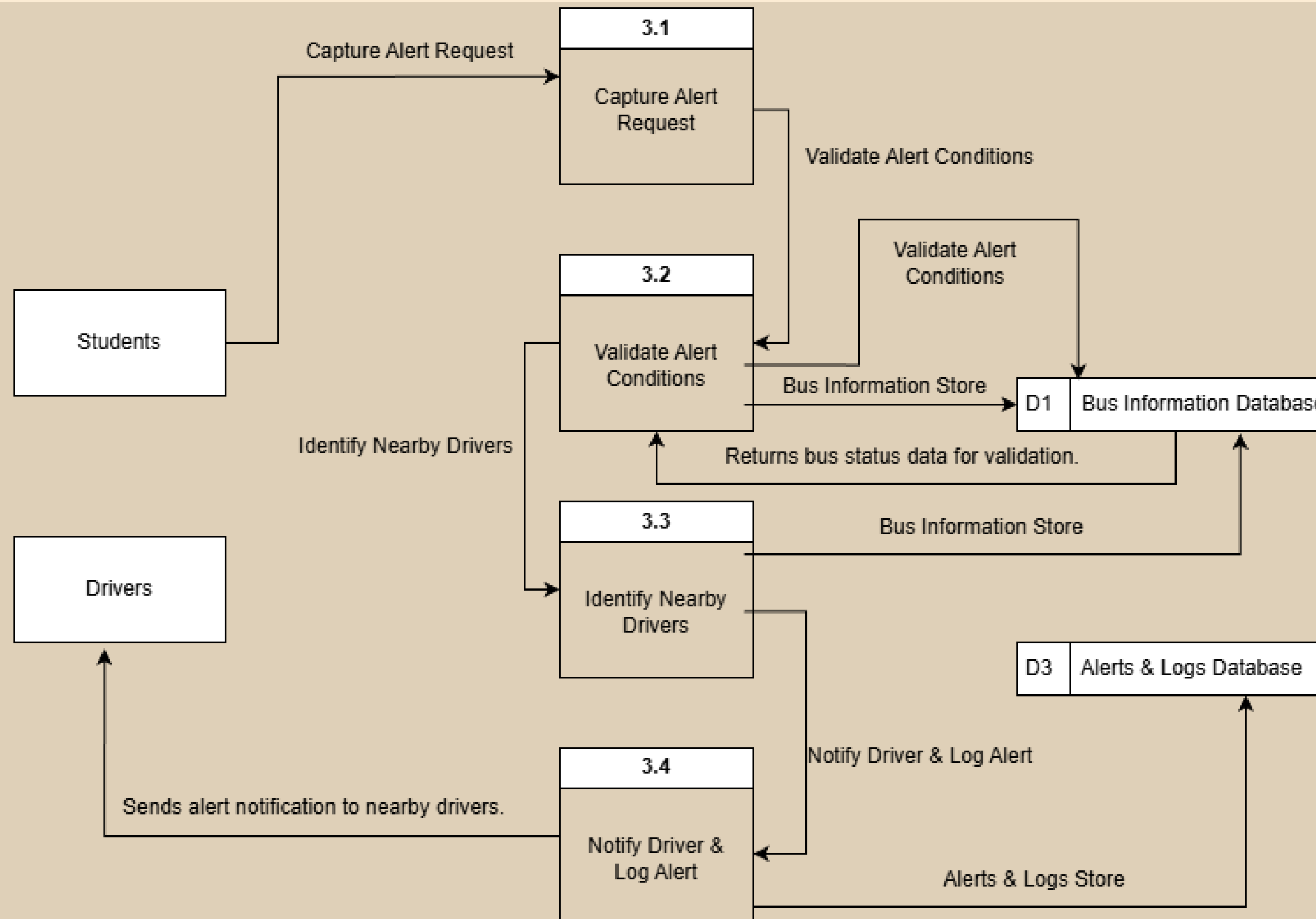
DATA FLOW DIAGRAM LEVEL 1



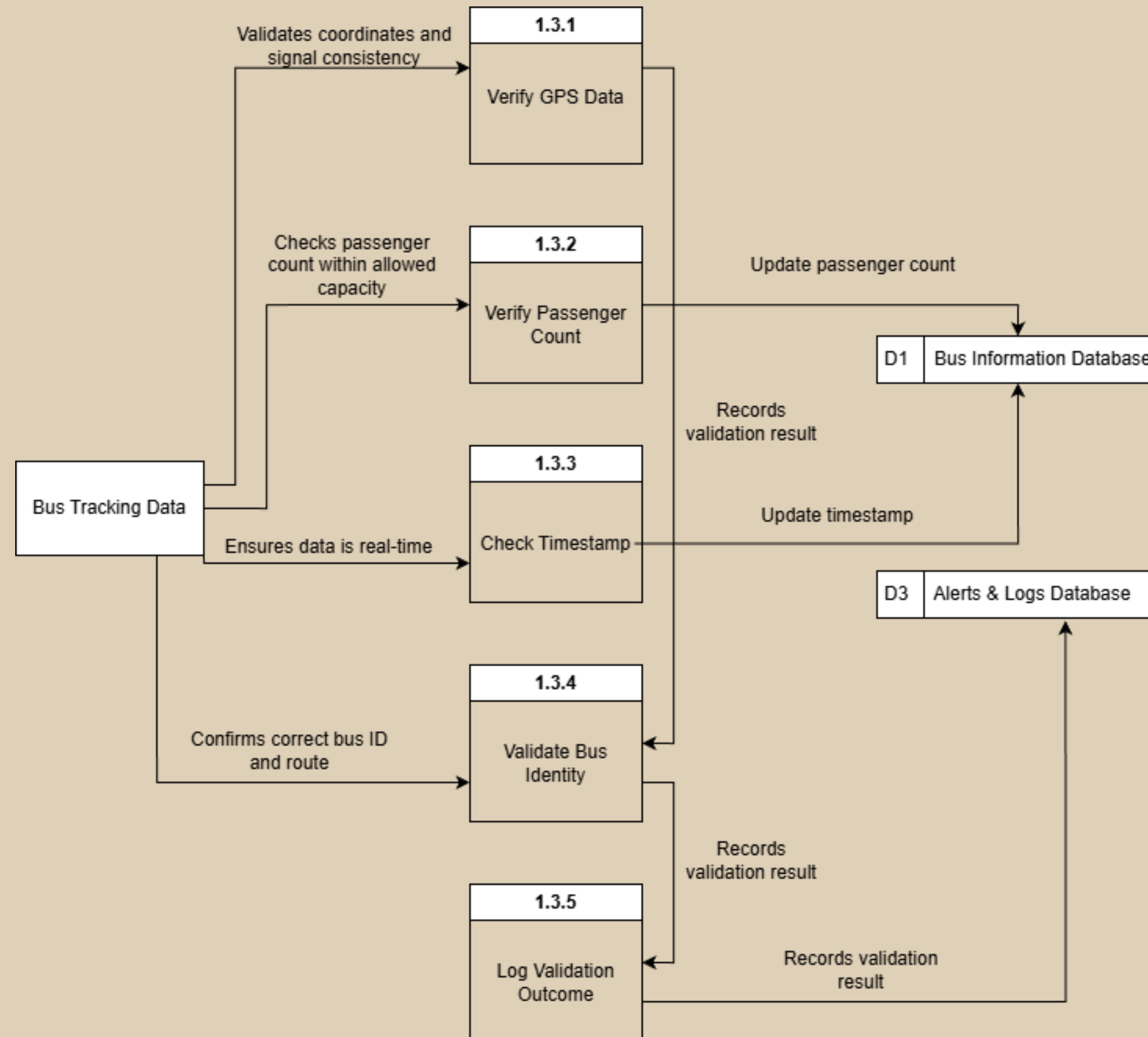
DATA FLOW DIAGRAM 2 BUS TRACKING HARDWARE



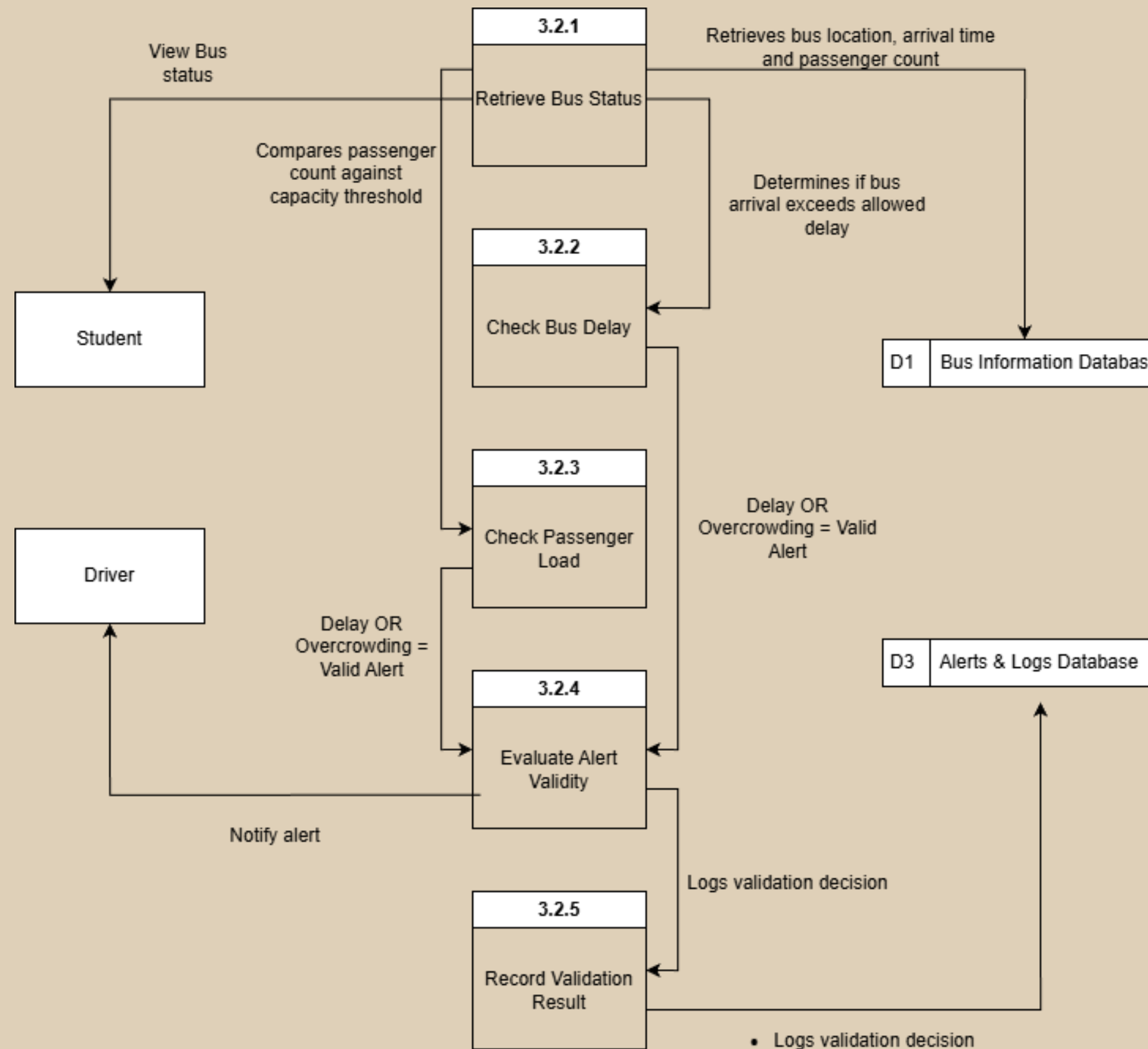
DATA FLOW DIAGRAM 2 STUDENTS & DRIVERS



DATA FLOW DIAGRAM₃ VALIDATE ALERT CONDITIONS

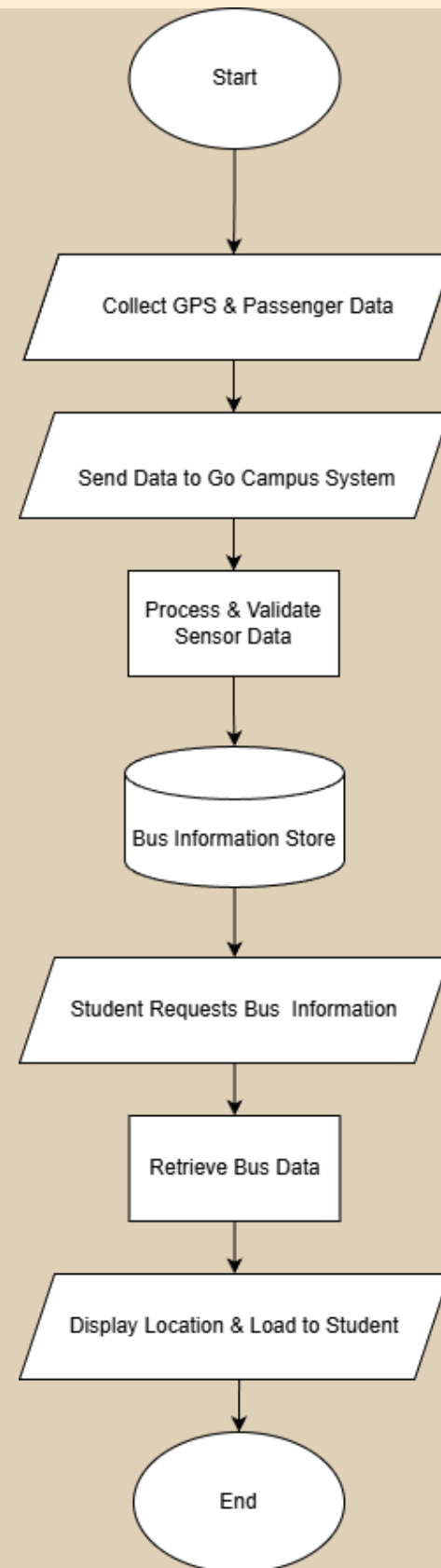


DATA FLOW DIAGRAM₃ VALIDATE ALERT CONDITIONS

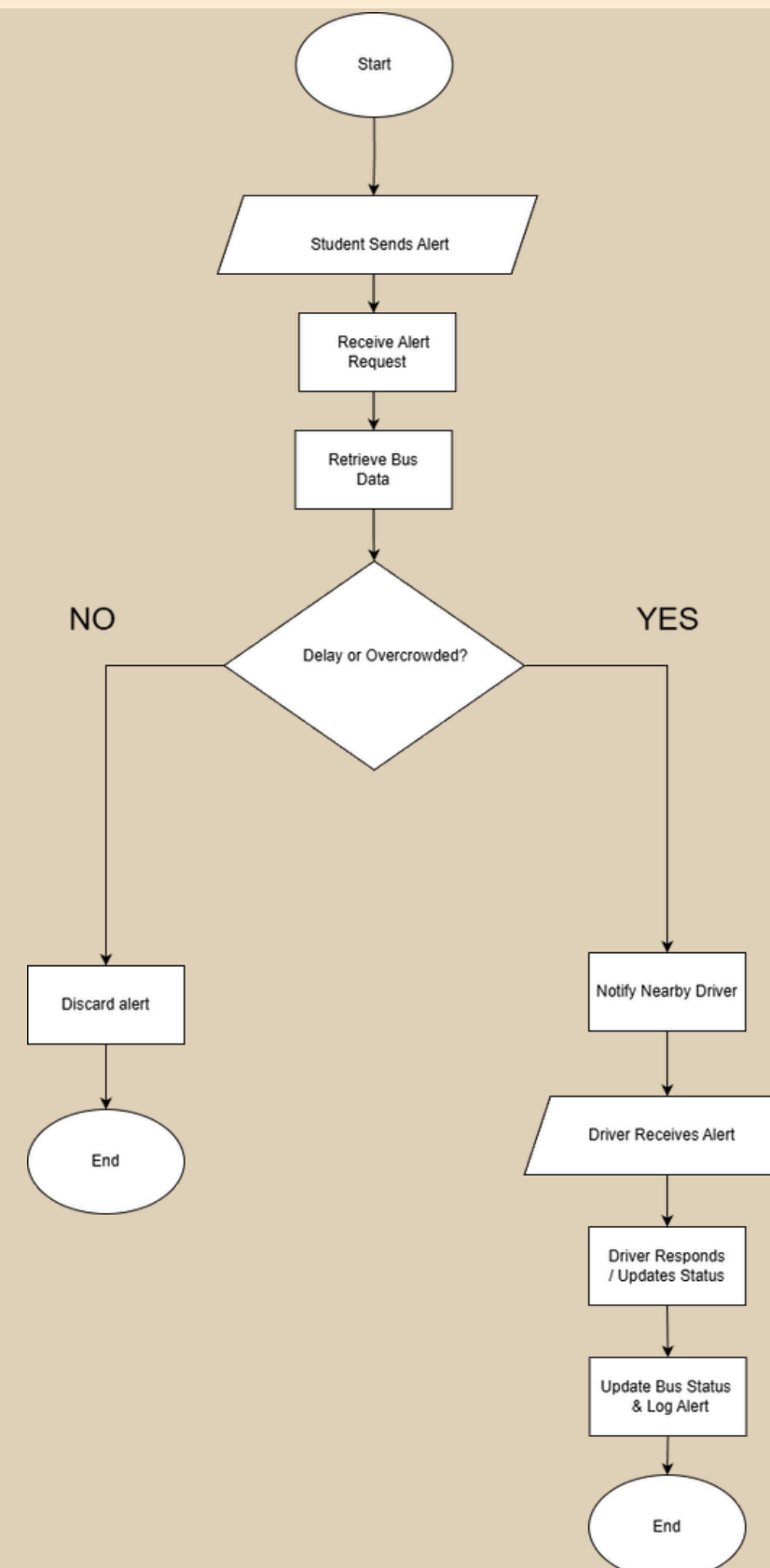


- Logs validation decision

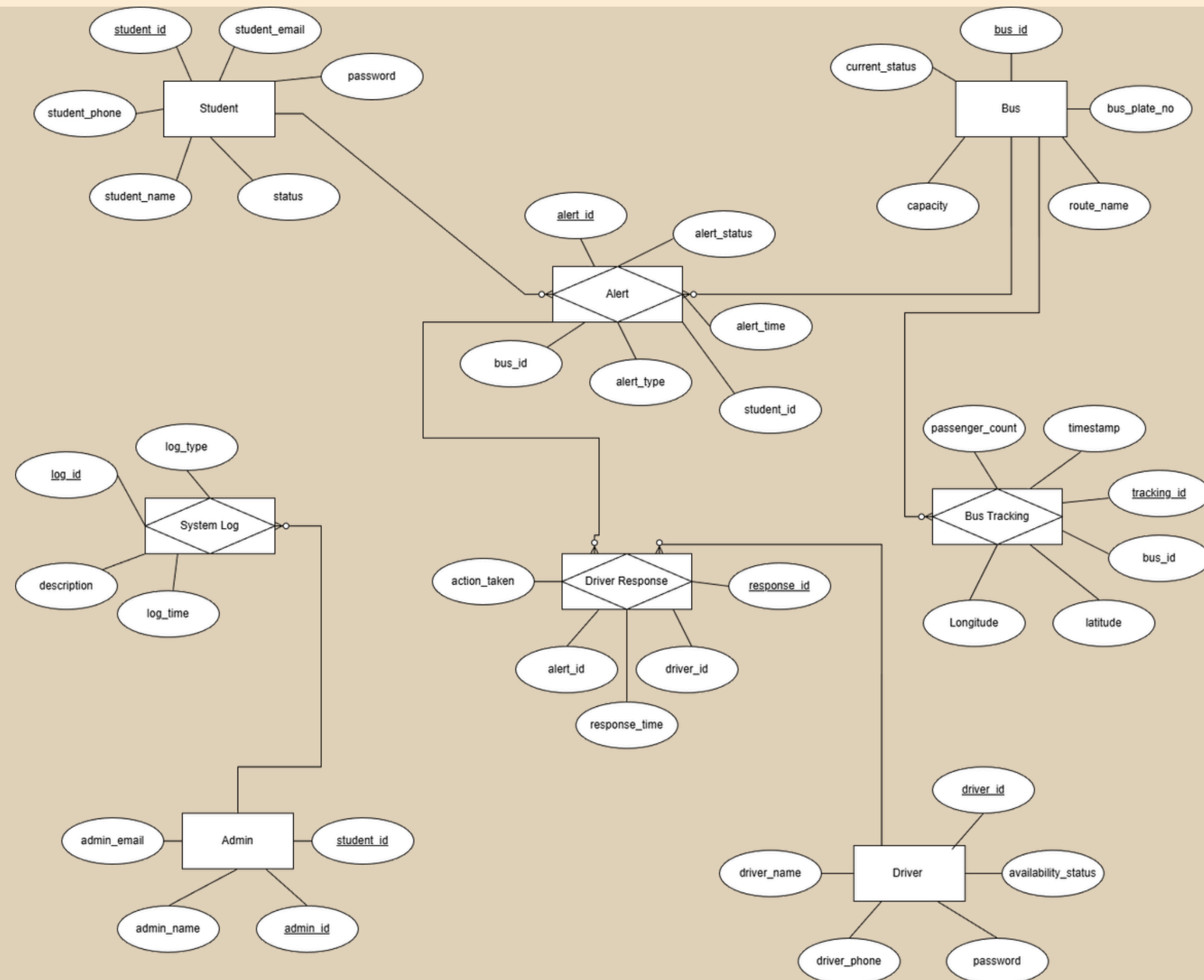
FLOWCHART BUS TRACKING AND STUDENT



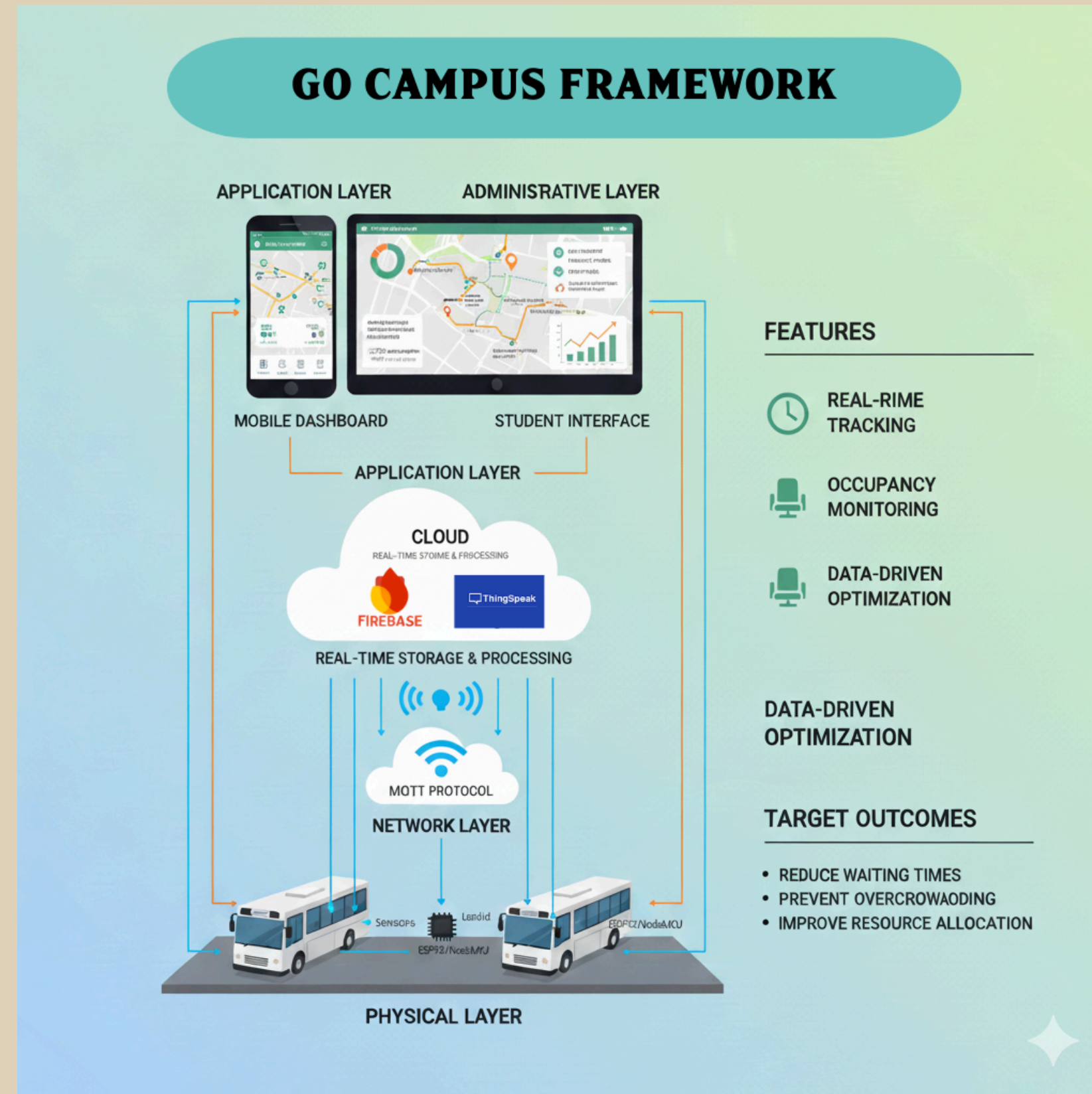
FLOWCHART STUDENT ALERT & DRIVER RESPONSE



ENTITY RELATIONSHIP DIAGRAM



FRAMEWORK



SOLUTION COMPLEXCITY

Real-Time Responsiveness	The algorithm processes live GPS and passenger count data instantly, allowing students to view accurate bus location and load information in real time.
Event-Driven Operation	System actions are triggered only when specific events occur, such as bus delays or overcrowding, ensuring efficient system performance.
Threshold-Based Decision Making	Predefined limits for passenger capacity and arrival time enable consistent and automatic alert generation without manual intervention.
Lightweight and Efficient	Rule-based logic requires minimal processing power, making it suitable for ESP32 hardware and mobile application environments.

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SOLUTION COMPLEXITY

The Go Campus system applies a **rule-based event-driven algorithm combined with threshold logic** to manage real-time shuttle monitoring and user interaction. The algorithm continuously receives data from IoT sensors, including GPS location and passenger count, and evaluates this data against predefined conditions such as maximum bus capacity and expected arrival time. When the incoming data satisfies specific rules—such as passenger count exceeding the capacity threshold or a delay beyond the allowed time window—the system automatically triggers corresponding events. These events include updating the bus status, notifying students in real time, and sending alerts to nearby drivers. This approach allows the system to respond immediately to changing conditions without requiring continuous manual input, ensuring efficient, real-time decision-making within the campus transportation environment.

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SOLUTION COMPLEXCITY

Scalability	New rules and conditions can be added easily to support additional buses, routes, or alert types without redesigning the system architecture.
Reliability	Fixed rules and thresholds ensure predictable and accurate system behavior, which is critical for campus transportation monitoring.
Ease of Validation	The clear and transparent logic allows supervisors and evaluators to easily understand, test, and validate the system behavior.
Commonly Used in IoT Systems	Rule-based event-driven algorithms are widely adopted in smart transportation and IoT applications, ensuring proven effectiveness.

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PROOF OF CONCEPT - STUDENT



Go Campus

Live shuttle tracking & alerts

☒ Student

☐ Driver

☐ Admin

Sign in

Register

Demo accounts preloaded:
admin/admin, Raven/123, Driver/123

Student

Welcome, Student

A

Next stop: DormsETA: 1 min

Passenger

Actions

Report Delay

Passenger Count

20:54 — Bus passed Library

20:54 — Bus passed Central Park

20:53 — Bus passed Central Park

20:53 — Bus passed Library

20:53 — Bus passed Library

Student

Welcome, Student

A

Report Delay

30 mins Delay

Cancel

Send

20:54 — Bus passed Central Park

20:54 — Bus passed Main Gate

20:54 — Bus passed Gate A

20:54 — Bus passed Library

20:54 — Bus passed Library

This page says

Current passenger count: 30 / 40

OK

A

Next stop: Central ParkETA: 1 min

Pass

Actions

Report Delay

Passenger Count

20:55 — Bus passed Gate A

20:55 — Bus passed Gate A

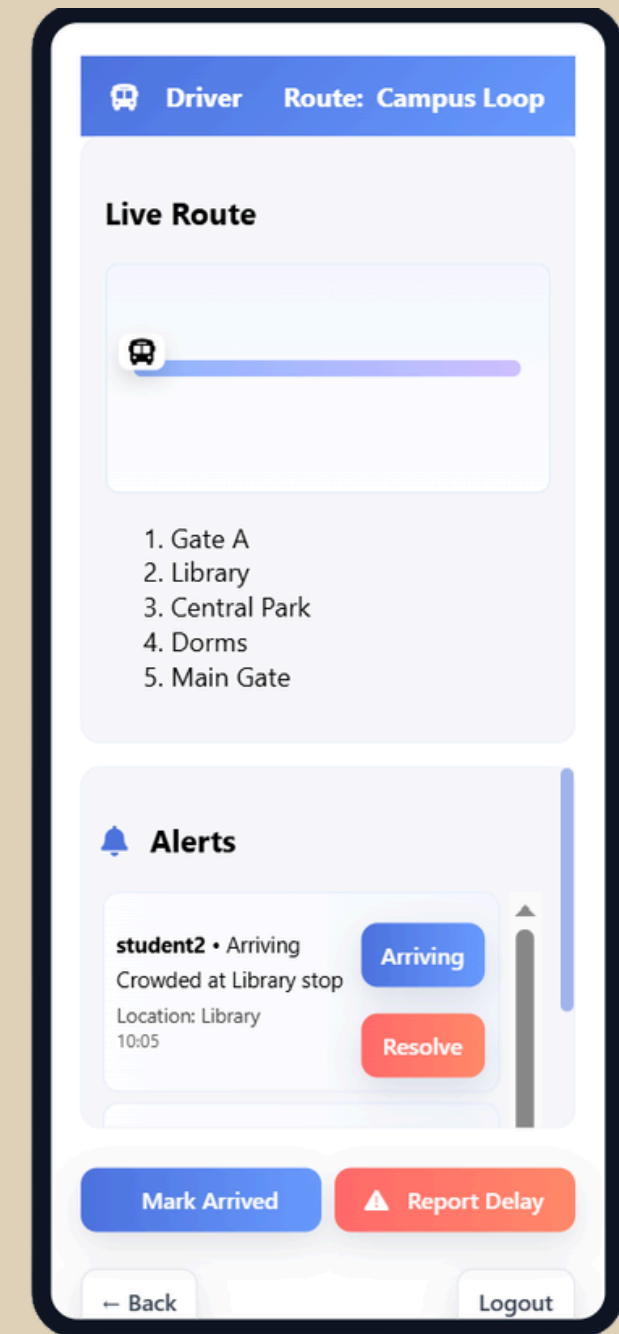
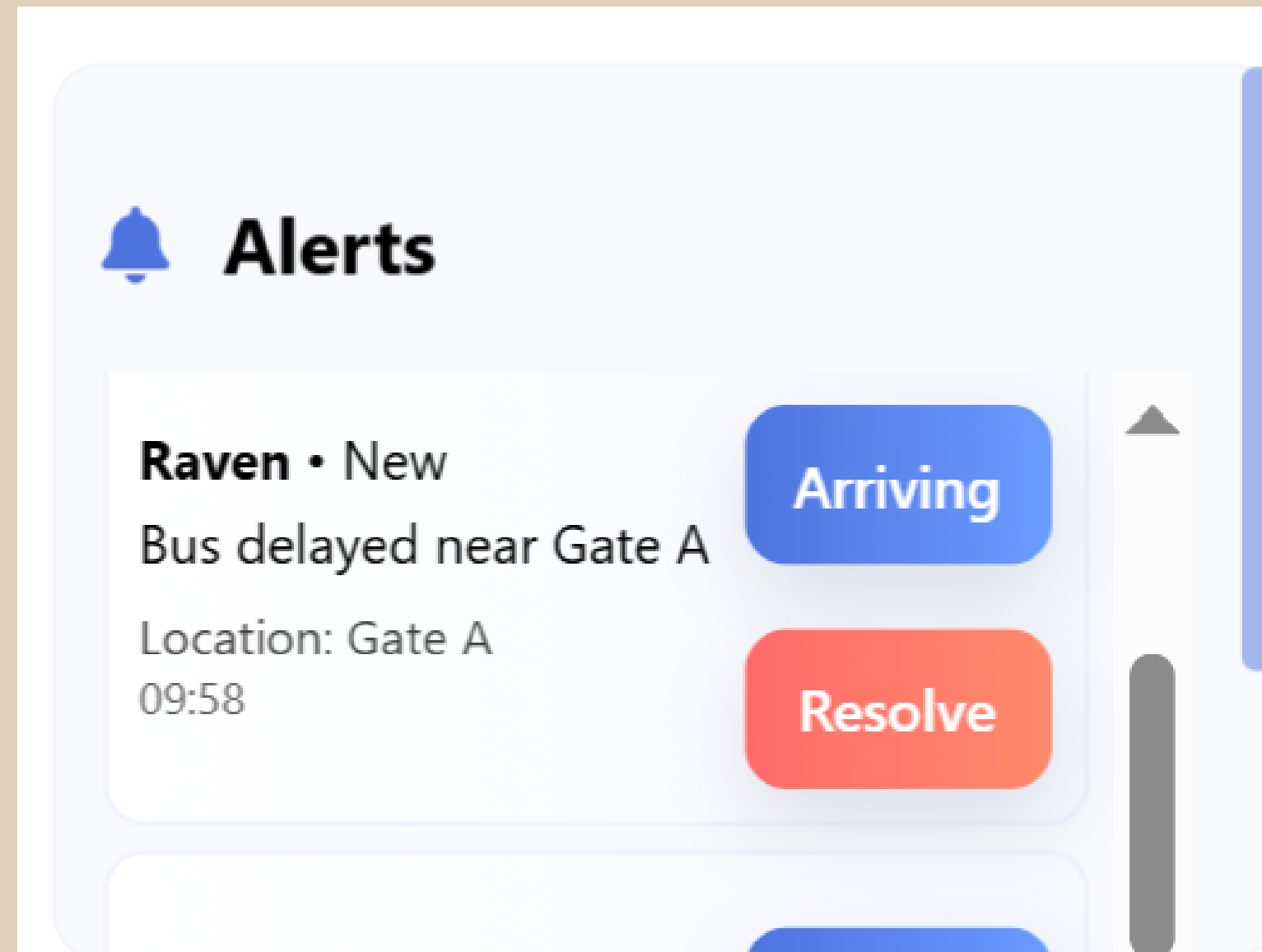
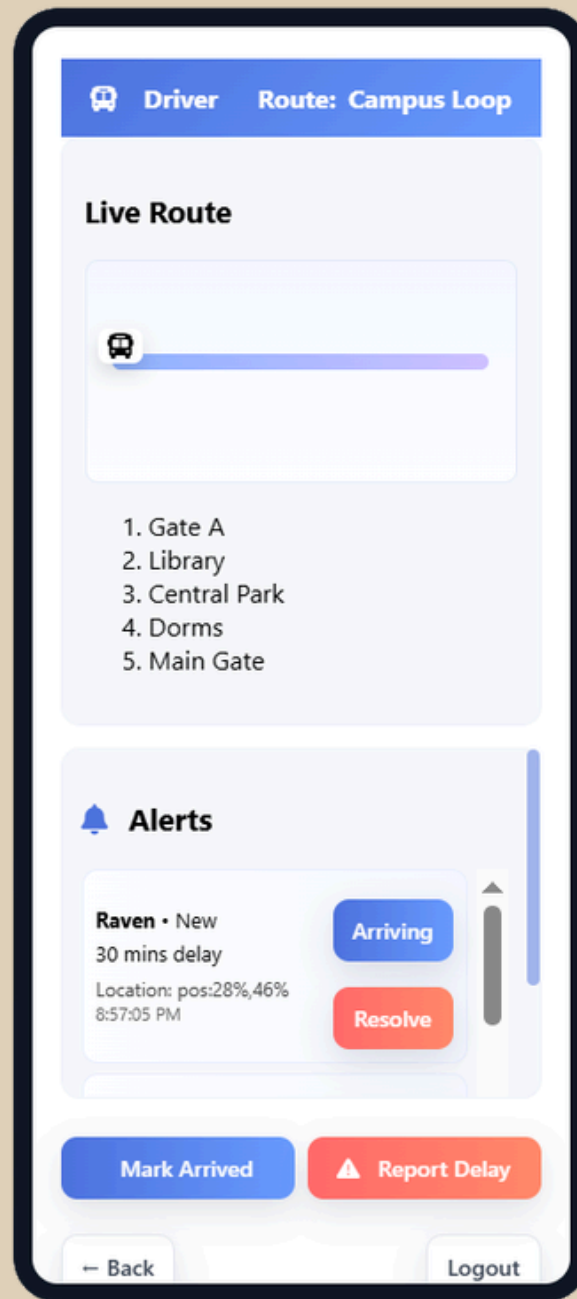
20:55 — Bus passed Central Park

20:55 — Bus passed Library

20:54 — Bus passed Central Park

Sent to nearby drivers

PROOF OF CONCEPT - DRIVER



PROOF OF CONCEPT - ADMIN

Admin

System Control

Users

New username

Password

Student

Create

Alerts (Admin)

student3 10:22

Bus skipped stop earlier

Location: Central Park

Driver Reply:

Mark Resolved

Delete

System Logs

[object Object]

Driver status: Arrived (8:58:20 PM)

[object Object]

[object Object]

[object Object]

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CONCLUSION

The Go-Campus: Dynamic Shuttle Tracking and Load Monitoring System successfully demonstrates how IoT technology can improve campus transportation efficiency. By integrating GPS tracking and passenger load monitoring sensors with a cloud-based dashboard, the system provides students with real-time information on shuttle location and seat availability. This reduces unnecessary waiting time, prevents overcrowding, and allows students to plan their travel more efficiently. At the same time, campus administrators benefit from access to accurate shuttle usage data, enabling better decision-making and route optimization. Overall, Go-Campus proves that IoT can contribute to a smarter, more convenient, and more efficient campus environment.



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THANK YOU

